ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R does not explain why the image is inverted.
2	**(c)** -25 cm	f = -(far point) for objects at infinity.
3	**(b)** smaller wavelengths are scattered more by small particles	Rayleigh type scattering.
4	**(c)** A is true but R is false.	A is true; R is false: the image forms in front of the retina.
5	**(b)** -1 D and +1.5 D	Distance: f = -100 cm. Reading: 1/f = -1/40 + 1/25 = 3/200, P = +1.5 D.
6	**(a)** blue light more strongly than red light	Shorter wavelengths are scattered more.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Higher n means more bending.
8	**(b)** the ciliary muscles changing the shape of the lens	Accommodation by the ciliary muscles.
9	**(a)** Both A and R are true and R is the correct explanation of A.	In myopia the image forms in front of the retina; the diverging lens shifts it back onto the retina.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R describes how the rainbow forms but does not explain the required geometry of Sun, observer and rain.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Red is scattered the least (1); The scattered light is dominated by shorter wavelengths (1).	1+1
12	Colours always separate (1); Because n differs for each colour (1).	1+1
13	Iris limits the entry of light (1); Lens changes its curvature to focus (1).	1+1
14	1/f = -1/50 + 1/25 = 1/50 (1); P = +2 D (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	1/f = -1/30 + 1/25 (1); 1/f = 1/150, f = 150 cm (1); P = +0.67 D (1).	1+1+1
16	Light reaches us after passing through a thick layer of air (1); Light is scattered many times by particles of all sizes (1); Colours mix and appear whitish (1).	1+1+1
17	At 10 cm (1); The lens must be thicker and have a shorter focal length (1); For 100 cm the lens is thinner (1).	1+1+1
	OR: Hypermetropia and myopia (1); Reading: 1/f = -1/100 + 1/25, P = +3 D (1); Distance: f = -2 m, P = -0.5 D (1).	1+1+1
18	Prism disperses the light (1); Edges appear coloured due to unequal deviations of colours (1); Violet and red are separated (1).	1+1+1
	OR: Sky is bright due to scattered sunlight (1); Stars' light is overpowered (1); At night no scattering (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Myopia (short-sightedness) (1). (ii) Concave (diverging) lens (1). (iii) Excessive curvature of the eye lens, or elongation of the eyeball (1). (iv) The lens must form the image of a distant object at the far point: $f = -80$ cm $= -0.8$ m (1); $P = \frac{1}{f} = \frac{1}{-0.8} = -1.25$ D (1).	1+1+1+2
20	(a) 1/f = -1/50 + 1/25 = 1/50 (1); P = +2 D (1); (b) f = 100 cm, 1/u = 1/v - 1/f = -1/50 - 1/100 (1); 1/u = -3/100 (1); u = -33.3 cm, so near point 33.3 cm (1).	1+1+1+1+1
	OR: (a) Red scatters least, so visible from far (1); (b) Yellow/red fog lamps penetrate better (1); (c) Earth sky is blue due to scattering (1); (c) Moon has no atmosphere, so a black sky (1); Scattering depends on wavelength and particle size (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a myopic eye is corrected with a convex lens, or hypermetropia with a concave lens; myopia needs concave and hypermetropia needs convex.
- Believing that most refraction in the eye occurs at the eye lens; it occurs at the outer surface of the cornea, and the lens only fine-tunes the focal length.
- Writing that the image on the retina is erect; it is real and inverted and the brain interprets it as erect.
- Stating that red light is scattered the most; it is scattered the least, and blue light is scattered the most.
- Forgetting the negative sign for a myopia lens, or using f in centimetres to find power; $P = 1/f$ with f in metres.
- Saying that planets twinkle like stars; stars twinkle due to atmospheric refraction of a point source, whereas planets appear as extended sources.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** white light	The second prism reverses the dispersion.
2	**(c)** go undeviated	No relative refraction.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Atmospheric refraction.
4	**(c)** is nearly monochromatic	Single wavelength.
5	**(a)** violet travels slower than red	n(violet) > n(red).
6	**(a)** Both A and R are true and R is the correct explanation of A.	Averaging of variations.
7	**(a)** Pass a beam through water with a few drops of milk	Colloidal milk particles scatter light.
8	**(b)** Light of all colours gets scattered many times	Multiple scattering mixes colours.
9	**(c)** A is true but R is false.	A hypermetropic eye has a short eyeball (or too long a focal length); the eyeball is too long in myopia. So R is false.
10	**(d)** A is false but R is true.	The sky is blue because blue light is scattered more; so A is false and R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distant rays are parallel and the eye can accommodate (1); Near objects need greater accommodation, which is lacking (1).	1+1
12	Hypermetropia (1); Near objects are focused behind the retina (1).	1+1
13	f = -60 cm (1); P = 1/(-0.6) = -1.67 D (1).	1+1
14	Their light passes through more atmosphere (1); Greater variation in the refraction (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Atmospheric particles are smaller than the wavelength (1); Short wavelengths scatter more, so noon sky is blue (1); Long path at sunset scatters away blue, leaving red (1).	1+1+1
16	Sun emits less violet (1); Upper atmosphere absorbs violet (1); Our eyes are less sensitive to violet (1).	1+1+1
17	Dark/black (1); Moon has no atmosphere (1); Without scattering the Sun's light isn't spread (1).	1+1+1
	OR: Red bends the least and returns at a larger angle (1); Violet bends the most and returns at a smaller angle (1); Hence red appears at the top (outer) and violet at the bottom (1).	1+1+1
18	Tiny droplets scatter the light (1); Scattering spreads light in all directions (1); Light seems to surround the source (1).	1+1+1
	OR: In the dark more light is needed (1); In bright light the pupil shrinks to protect the retina (1); Controlled by the iris muscles (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Dispersion (1). (ii) Red the least, violet the most (1). (iii) Spectrum (VIBGYOR) (1). (iv) Tiny water droplets act as small prisms: sunlight is refracted and dispersed, internally reflected inside the drop and refracted again as it comes out (1); different colours reach the eye of the observer from different droplets to form the rainbow (1).	1+1+1+2
20	Refraction as light enters the drop (1); Dispersion since n varies with colour (1); Internal reflection at the back (1); Sun must be behind the observer, low in the sky (1); Red bends least and lies at the top (1).	1+1+1+1+1
	OR: In dim light the iris dilates the pupil (1); In bright light it contracts the pupil (1); For near objects the ciliary muscles contract (1); Lens thickens, focal length decreases (1); For distant objects the opposite (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a myopic eye is corrected with a convex lens, or hypermetropia with a concave lens; myopia needs concave and hypermetropia needs convex.
- Believing that most refraction in the eye occurs at the eye lens; it occurs at the outer surface of the cornea, and the lens only fine-tunes the focal length.
- Writing that the image on the retina is erect; it is real and inverted and the brain interprets it as erect.
- Stating that red light is scattered the most; it is scattered the least, and blue light is scattered the most.
- Forgetting the negative sign for a myopia lens, or using f in centimetres to find power; $P = 1/f$ with f in metres.
- Saying that planets twinkle like stars; stars twinkle due to atmospheric refraction of a point source, whereas planets appear as extended sources.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** slightly less	No advance sunrise or delayed sunset.
2	**(a)** Both A and R are true and R is the correct explanation of A.	No scattering in space.
3	**(b)** it is much closer and extended	Averaging effect.
4	**(d)** the far point of the eye is 50 cm	f = -50 cm.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Presbyopia with myopia.
6	**(d)** bifocal lenses	Two corrections are needed.
7	**(c)** about 67 cm	1/v = 1/40 - 1/25 = -3/200.
8	**(a)** Both A and R are true and R is the correct explanation of A.	The observer must stand between the Sun and the rain to receive the returned light.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The image falls short of the retina, so the distant object looks blurred.
10	**(c)** A is true but R is false.	Red has the longest wavelength and is scattered the least, so it is visible from far; R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	f = -1/3 m = -33.3 cm (1); Far point = 33.3 cm (1).	1+1
12	n is larger for violet (1); Violet travels slower in glass (1).	1+1
13	Light is monochromatic (1); Only one wavelength exists (1).	1+1
14	Slab faces are parallel (1); Prism faces are inclined, so the dispersion adds up (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Light refracts at the first surface and disperses (1); It reflects once inside at the back surface (1); It refracts again as it exits, with violet deviated the most (1).	1+1+1
16	Clear air has small molecules scattering mainly blue (1); Dust scatters all colours (1); So the dusty sky looks whitish (1).	1+1+1
17	Concave lens makes objects look smaller (1); Convex lens makes objects look bigger (1); Concave helps far vision (1).	1+1+1
	OR: v = 2 cm, u = -25 cm (1); 1/f = 1/2 + 1/25 = 27/50, f = 1.85 cm (1); Possible only if the lens can shorten its focal length to 1.85 cm (1).	1+1+1
18	The lens itself becomes cloudy (1); Spectacles cannot clear it (1); Surgery replaces the lens (1).	1+1+1
	OR: Rods work in dim light (1); Cones work in bright light and detect colour (1); Both convert light into electrical signals (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Scattering of light (1). (ii) Blue light has a shorter wavelength than red and is scattered much more strongly by atmospheric particles (1). (iii) Red light is scattered the least by fog or smoke, so it can be seen from far (1). (iv) Near the horizon sunlight travels a longer path through the atmosphere, so most blue light is scattered away and red reaches the eye (1); there is no atmosphere in space to scatter sunlight, so the sky looks dark (1).	1+1+1+2
20	Longer path scatters blue away so red remains (1); Rays from the lower and upper edges are refracted differently (1); The disc looks flattened (1); Atmospheric refraction bends rays over the horizon (1); Sun is seen about 2 minutes after the actual sunset (1).	1+1+1+1+1

	OR: Myopia: f = -125 cm (1); P = -0.8 D, concave (1); Hypermetropia: 1/f = -1/100 + 1/25 = 3/100 (1); f = 33.3 cm, P = +3 D, convex (1); Concave diverges for myopia; convex converges for hypermetropia (1).	1+1+1+1+1

Mistakes to avoid:

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